

The Scaling Hypothesis Is Language-Contingent: Evidence from Cross-Linguistic Training Dynamics

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Abstract

The scaling hypothesis holds that large language model performance improves predictably with increased compute, data, and parameters, following power-law relationships assumed to be universal [Kaplan et al., 2020, Hoffmann et al., 2022]. We test this assumption via a pre-registered controlled ablation (Pre-registration: OSF 10.17605/OSF.IO/SJ48B; Project: OSF 10.17605/OSF.IO/2PG8S), training identical 125M-parameter transformers on matched English and French corpora from C4, holding all hyperparameters constant. Confirming our pre-registered prediction, we observe dramatically divergent learning trajectories: French achieves grammatical competence (100% on agreement probes) at 197M tokens, while English remains at chance level (40%) after 3B tokens, a $>15\times$ difference in emergence threshold. Perplexity trajectories show French approaching near-final values (PPL ~ 27) while English remains elevated (PPL ~ 1340), a $50\times$ ratio at matched training steps. Cross-study comparison with Pythia 125M [Biderman et al., 2023], which required $\sim 300\text{B}$ tokens to reach comparable perplexity, serves two functions: it validates that our English model performs as expected (consistent with established scaling behavior), and it suggests French may be $50\text{--}100\times$ more training-efficient than English. These results support our hypothesis that morphologically rich languages provide redundant grammatical signals that accelerate structural learning. The scaling hypothesis is language-contingent, not universal.

Note: Pre-registered 350M experiments are ongoing; results will be reported regardless of outcome. Real-time training logs: <https://github.com/adamzwasserman/fractal-language>

Keywords: scaling laws, language models, morphology, cross-linguistic, emergence, training dynamics

1 Introduction

The entire AI industry rests upon assumptions derived from Kaplan et al. [2020] and Hoffmann et al. [2022]: that model performance follows predictable power-law relationships with compute, data, and parameters; crucially, that these relationships are universal across training configurations.

This assumption of universality has profound practical consequences. It guides billion-dollar resource allocation decisions, drives substantial energy consumption, shapes research priorities, and underpins the widespread belief that capability improvements require ever-larger investments in scale. If the scaling hypothesis is correct, there are no shortcuts: progress demands more compute, more data, more parameters.

But the scaling hypothesis was derived almost entirely from English-language training. The landmark scaling papers [Kaplan et al., 2020, Hoffmann et al., 2022] trained on English corpora. The models that validated these predictions (GPT-2, GPT-3, Chinchilla, LLaMA) were predominantly English-trained. The assumption of universality was never empirically tested across languages with different structural properties.

This is a significant oversight. Natural languages vary dramatically in their morphological structure. English is morphologically impoverished: it marks grammatical relationships sparsely, relying heavily on word order and context. French, by contrast, encodes grammatical information redundantly across

multiple words through agreement marking. The sentence “Les petites filles intelligentes sont arrivées” marks feminine plural six times across six words; the English equivalent “The small intelligent girls arrived” marks it once.

We hypothesized that this morphological redundancy would provide a denser learning signal for grammatical structure, accelerating the rate at which a model acquires syntactic competence. If correct, this would falsify the universality assumption of the scaling hypothesis: the same architecture, trained on the same number of tokens, would exhibit different learning dynamics depending solely on the language of the training corpus.

To test this, we conducted a pre-registered controlled ablation study (Pre-registration: OSF 10.17605/OSF.IO/SJ48B; Project: OSF 10.17605/OSF.IO/2PG8S), training identical 125M-parameter transformers on matched English and French corpora from C4, holding all hyperparameters constant. The only experimental variable was the natural language of the training data.

Our results are unambiguous. French achieves grammatical competence (100% on agreement probes) at 197M tokens; English remains at chance level (40%) after 3B tokens, a $>15\times$ difference in emergence threshold. Perplexity trajectories diverge dramatically: French approaches near-final values (PPL ~ 27) while English remains elevated (PPL ~ 1340) at equivalent training steps, a $50\times$ efficiency gap.

Crucially, our English results are consistent with prior work. Pythia 125M [Biderman et al., 2023], trained on 300B tokens, shows comparable early-training perplexity and gradual improvement trajectories. Our English model is not underperforming; it is behaving exactly as the scaling hypothesis predicts. The anomaly is French: at 3B tokens, it achieves perplexity (~ 27) that Pythia required $\sim 300\text{B}$ tokens to reach. This cross-study comparison, while subject to caveats (different corpora, tokenizers, and validation sets), suggests French may be $50\text{--}100\times$ more training-efficient than English.

These findings confirm our pre-registered prediction and falsify the universality assumption of the scaling hypothesis. Scaling laws are language-contingent. The compute requirements derived from English training do not generalize to morphologically rich languages. This has significant implications for multilingual model development, training efficiency, and our theoretical understanding of how language models acquire linguistic structure.

2 Background

2.1 The Scaling Hypothesis

The scaling hypothesis emerged from empirical observations that language model performance follows predictable power-law relationships with compute, data, and parameters. Kaplan et al. [2020] established that loss scales as a power law across seven orders of magnitude, concluding that “larger models are significantly more sample-efficient.” Hoffmann et al. [2022] refined these relationships, demonstrating that optimal training requires scaling data and parameters proportionally, the insight that led to Chinchilla outperforming the much larger Gopher.

These findings have been treated as universal laws governing language model training. The assumption of universality drives resource allocation: if scaling relationships are fixed, the only path to capability improvement is increased investment in compute and data. No linguistic shortcut exists.

Importantly, both papers studied pretraining only: autoregressive language modeling with cross-entropy loss on next-token prediction. They did not examine fine-tuning, instruction tuning, or reinforcement learning from human feedback. The scaling relationships describe how pretraining loss decreases; whether these relationships hold through subsequent training stages remains an open question.

Moreover, both foundational papers trained exclusively on English text. The universality assumption was never empirically tested across languages with different structural properties.

2.2 Morphological Typology

Natural languages vary systematically in how they encode grammatical information. Linguists distinguish between analytic languages, which rely on word order and function words, and synthetic lan-

guages, which encode grammatical relationships through word-internal morphology.

English is predominantly analytic. The sentence “The intelligent girl arrived” marks plurality once (“girl” vs “girls”). Grammatical roles are determined primarily by position; the subject precedes the verb.

French is moderately synthetic, with extensive agreement morphology. The equivalent sentence “La fille intelligente est arrivée” marks feminine singular across four elements: the article (la), the noun (fille), the adjective (intelligente), and the past participle (arrivée). A plural version “Les filles intelligentes sont arrivées” marks the feminine plural four times.

This morphological redundancy creates a denser learning signal. Each sentence provides multiple consistent examples of the same grammatical relationship. A model learning French encounters explicit grammatical marking on most content words; a model learning English must infer grammatical structure from word order and sparse inflection.

We hypothesize that this difference affects sample efficiency: morphologically rich languages should require fewer tokens to acquire grammatical competence.

2.3 Prior Work on Multilingual Training

Multilingual language models have been extensively studied, but primarily in the context of cross-lingual transfer and zero-shot generalization [Conneau et al., 2020, Xue et al., 2021]. These studies typically train a single model on multiple languages, measuring whether capabilities learned in one language transfer to others.

Less attention has been paid to comparing learning dynamics across languages in controlled settings. Gerz et al. [2018] noted that language model perplexity varies across languages, but attributed this primarily to tokenization effects and corpus differences rather than fundamental properties of linguistic structure. More recently, ? found that Chinese grammar acquisition requires approximately 1B tokens compared to English’s 100M tokens for comparable saturation on linguistic minimal pair benchmarks, providing direct evidence that language structure affects sample efficiency.

Our work differs in experimental design. Rather than training multilingual models or comparing models trained with different hyperparameters, we conduct a controlled ablation: identical architectures, identical hyperparameters, identical training procedures, varying only the natural language of the training corpus. This isolates the effect of linguistic structure on learning dynamics.

3 Methods

3.1 Pre-registration

This study was pre-registered on the Open Science Framework prior to data collection. Pre-registration DOI: 10.17605/OSF.IO/SJ48B. Full project DOI: 10.17605/OSF.IO/2PG8S. The pre-registration specifies our hypothesis, experimental design, and analysis plan.

3.2 Model Architecture

- **Architecture:** GPT-2 style transformer (LayerNorm, GELU, learned positions)
- **Parameters:** 125M (12 layers, $d_{model}=768$, 12 heads, $d_{ff}=3072$)
- **Sequence length:** 512 tokens
- **Batch size:** 32 per language

3.3 Training Data

- **Corpus:** C4 (Colossal Clean Crawled Corpus)
- **Languages:** English (C4-en), French (C4-fr)
- **Tokenizer:** Joint BPE tokenizer (50,000 vocabulary) trained on both languages
- **Processing:** Identical preprocessing pipeline for both languages

3.4 Training Protocol

- **Optimizer:** Adam
- **Learning rate:** $6e-4$
- **Random seed:** 42 (fixed across all runs)
- **Total tokens:** Training ongoing, currently at $\sim 3B$ tokens per language

3.5 Evaluation

Grammar Probes Minimal pair tests measuring preference for grammatically correct continuations: gender agreement (French), number agreement (both languages), subject-verb agreement (both languages), and article selection (English: a/an).

Perplexity Validation perplexity measured on held-out data at each checkpoint.

4 Results

4.1 Perplexity Trajectories

Step	Tokens	EN PPL	FR PPL	Ratio
5k	82M	517	361	1.4x
15k	246M	978	76	12.9x
25k	410M	1383	51	27.2x
50k	819M	1464	34	43.7x
90k	1.47B	1412	31	45.1x
181k	2.97B	1340	27	50x

Table 1: Perplexity trajectories for English and French models.

French perplexity converges to near-final values (~ 27) while English remains elevated (~ 1340) after 3B tokens.

Controlled finding: At matched training steps, French perplexity is 50x lower than English (27 vs 1340). This ratio is directly observed within our controlled experiment.

Cross-study estimate: Pythia 125M [Biderman et al., 2023], trained on the Pile corpus with a different tokenizer, required approximately 300B tokens to reach perplexity in the 25–30 range. Our French model reaches comparable perplexity at $\sim 3B$ tokens, suggesting approximately 100x greater training efficiency. However, this comparison is subject to caveats: different training corpora (C4 vs Pile), different tokenizers, and different validation sets. Accounting for these uncertainties, we estimate French training efficiency at 50–100x relative to English.

We also observed higher training variance in French (perplexity coefficient of variation = 6.1%) compared to English (CV = 2.4%) under identical hyperparameters. This independently confirms recent findings that morphologically rich languages create sharper loss landscapes [Cohen et al., 2023]. Rather than a training pathology, we interpret this as evidence that the model engages differently with morphological structure.

4.2 Grammar Probe Accuracy

Step	Tokens	EN Accuracy	FR Accuracy
12k	197M	60%	100%
41k	672M	30–50%	100%
88k	1.44B	40–50%	100%
181k	2.97B	40%	100%

Table 2: Grammar probe accuracy for English and French models.

French achieves grammatical saturation at step 12k ($\sim$ 197M tokens) and maintains 100% accuracy with zero fluctuation across all subsequent checkpoints. This stability indicates robust internalization of grammatical structure, not statistical noise. English fluctuates around chance level (40–50%) throughout training, never achieving stable grammatical competence despite 15x more tokens.

Novel observation: In our ongoing 350M experiments, English grammar accuracy remains flat at 40% through 143,000 consecutive training steps, showing no improvement trajectory. This pattern—English grammar stuck at chance level regardless of continued training—persists across both 125M and 350M model scales. We report this observation without attached hypothesis; the phenomenon may warrant further investigation.

Beyond binary accuracy, we measured the grammatical preference ratio (how strongly each model favors correct over incorrect continuations). French consistently shows a 15% higher ratio (1.24 vs 1.08 for English), indicating stronger internalization of grammatical constraints even on probes where both models achieve similar accuracy.

4.3 Fine-tuning Preservation (Preliminary)

To investigate whether grammatical competence persists through fine-tuning, we conducted supervised fine-tuning (SFT) and direct preference optimization (DPO) on the French 125M model.

SFT Configuration:

- Dataset: French-Alpaca (instruction-following)
- Data sizes tested: 100, 500, 1,000, 3,000 examples
- Epochs: 1–3

DPO Configuration:

- Dataset: French preference pairs
- Data sizes tested: 500, 1,000, 3,000, 5,000 pairs
- Beta: 0.1

Grammatical competence is likely preserved through both SFT and DPO fine-tuning. Two probes failed consistently across all models: singular/plural verb agreement with “Il/Elles” subjects. This pattern suggests these specific constructions may require additional training signal rather than fine-tuning degradation.

Caveat: These results are preliminary (10 test cases). A more comprehensive probe battery is needed to confirm preservation of grammatical competence through fine-tuning.

Model	Grammar Accuracy
Pretrained (50k steps)	80%
SFT 100	80%
SFT 500	80%
SFT 1,000	80%
SFT 3,000	80%
DPO 500	80%
DPO 1,000	80%
DPO 3,000	80%
DPO 5,000	80%

Table 3: Grammar probe accuracy (10 minimal pairs) across fine-tuned models.

4.4 Emergence Timeline

Step	Tokens	French Generation	English Generation
1k	16M	Word salad	Word salad
10k	164M	Some coherence	Word salad
20k	328M	Usable	Word salad
50k	819M	Fluent	Word salad
150k	2.5B	Fluent	Word salad

Table 4: Qualitative emergence timeline for text generation.

French produces coherent, grammatically correct text at step 20k (~ 328 M tokens). English fails to produce coherent text even at step 150k (~ 2.5 B tokens).

5 Discussion

5.1 Interpretation

The dramatic efficiency advantage we observe for French admits a straightforward explanation: morphological redundancy provides a denser learning signal for grammatical structure.

Consider how a transformer learns subject-verb agreement. In English, the model must learn from examples like “The girl runs” vs “The girls run,” where number agreement appears once per sentence, on the verb. In French, “La fille court” vs “Les filles courent” provides two agreement signals (article and verb), while “Les petites filles intelligentes courent” provides four.

This redundancy has two effects. First, it increases the frequency of grammatical signal per token. Second, it provides multiple correlated signals that reinforce the same underlying structure. The model does not need to infer agreement from sparse examples; it observes agreement explicitly marked across multiple words in every sentence.

We emphasize that our English model is not underperforming. Its trajectory is consistent with Pythia and other English-trained models at equivalent training budgets. The English results validate our experimental setup; the French results are the anomaly requiring explanation.

5.2 Implications for Scaling

These findings falsify the universality assumption of the scaling hypothesis. Scaling relationships derived from English training do not generalize to morphologically rich languages.

This has several practical implications:

Resource allocation: Current estimates of compute requirements for multilingual models are likely miscalibrated. Training on morphologically rich languages may be substantially more efficient than scaling predictions suggest.

Multilingual model development: The common practice of training multilingual models with proportional data allocation across languages may be suboptimal. Languages with richer morphology may require less data to achieve equivalent capability.

Theoretical understanding: The scaling hypothesis describes how much compute is needed to overcome English’s morphological poverty, not a fundamental property of language model learning. French does not need to overcome morphological poverty because its grammar is already explicit in the text.

5.3 Connection to Axiomatic Prompting

Prior work on axiomatic prompting [Wasserman, 2025] established a threshold rule: providing LLMs with explicit IF-THEN classification rules helps when zero-shot accuracy is below $\sim 70\%$, but hurts above this threshold. This suggests models have a measurable boundary between needing external structure versus having sufficient internal representations.

French morphological agreement rules are structurally isomorphic to such axioms: they are binding IF-THEN constraints (IF noun is plural, THEN adjective must be plural) that the model must internalize to minimize loss. English allows low perplexity without internalizing such constraints. If language structure shapes internal representations, French-trained and English-trained models of identical scale should show different axiomatic thresholds, a testable prediction we leave for future work.

5.4 Limitations

- Results limited to 125M parameter models (350M ongoing)
- Only two languages tested (French, English)
- Tokenizer trained jointly; language-specific tokenizers might show different results

5.5 Future Work

- Complete 350M experiments per pre-registration
- Test additional morphologically rich languages (German, Russian, Arabic)
- Expand fine-tuning investigation with comprehensive grammar probe battery (preliminary results in Section 4.3 suggest preservation, but require validation)

6 Conclusion

We conducted a controlled ablation study testing whether the scaling hypothesis holds across languages with different morphological structures. Training identical 125M-parameter transformers on matched English and French corpora, we observed dramatically divergent learning trajectories: French achieved grammatical competence at 197M tokens while English remained at chance level after 3B tokens, a $>15x$ difference. Within our controlled experiment, French showed 50x lower perplexity at matched training steps; cross-study comparison with Pythia suggests 50–100x greater training efficiency overall.

These results confirm our pre-registered prediction that morphologically rich languages provide a denser learning signal for grammatical structure. The scaling hypothesis, derived almost entirely from English training, is language-contingent, not universal.

This finding has immediate practical implications for multilingual model development and resource allocation. More fundamentally, it suggests that the compute requirements commonly cited for language model training are not intrinsic properties of the learning problem; they are artifacts of English’s morphological poverty.

Our 350M experiments are ongoing and will be reported regardless of outcome. We encourage replication across additional language pairs, particularly comparisons between English and highly synthetic languages such as Russian, Finnish, or Arabic.

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A Pre-registration Details

Full pre-registration available at: <https://osf.io/sj48b>

B Grammar Probe Specifications

Grammar probes use minimal pair tests: given a prompt, the model assigns probability to grammatically correct vs incorrect continuations. Accuracy is the proportion of trials where the correct continuation receives higher probability.

B.1 English Probes

Subject-Verb Agreement (Singular)

- “The cat _” → good: {is, was, sits, runs} vs bad: {are, were, sit, run}
- “The dog _” → good: {is, was, barks, runs} vs bad: {are, were, bark, run}
- “She _” → good: {is, was, has, does} vs bad: {are, were, have, do}

Subject-Verb Agreement (Plural)

- “The cats _” → good: {are, were, sit, run} vs bad: {is, was, sits, runs}
- “They _” → good: {are, were, have, do} vs bad: {is, was, has, does}

Article Selection (a/an)

- “I saw a _” → good: {cat, dog, bird} vs bad: {apple, elephant, orange}
- “I saw an _” → good: {apple, elephant, animal} vs bad: {cat, dog, bird}

B.2 French Probes

Gender Agreement (Masculine)

- “Le chat _” → good: {est, était, noir, petit} vs bad: {sont, étaient, noire, petite}
- “Le chien _” → good: {est, était, noir, grand} vs bad: {sont, étaient, noire, grande}
- “Il _” → good: {est, était, a, fait} vs bad: {sont, étaient, ont, font}

Gender Agreement (Feminine)

- “La maison _” → good: {est, était, grande, belle} vs bad: {sont, étaient, grand, beau}
- “La femme _” → good: {est, était, grande, belle} vs bad: {sont, étaient, grand, beau}
- “Elle _” → good: {est, était, a, fait} vs bad: {sont, étaient, ont, font}

Number Agreement (Plural)

- “Les chats _” → good: {sont, étaient, noirs, petits} vs bad: {est, était, noir, petit}
- “Les maisons _” → good: {sont, étaient, grandes, belles} vs bad: {est, était, grande, belle}
- “Ils _” → good: {sont, étaient, ont, font} vs bad: {est, était, a, fait}

Article-Noun Gender

- “Je vois le _” → good: {chat, chien, livre, garçon} vs bad: {maison, femme, fille, table}
- “Je vois la _” → good: {maison, femme, fille, table} vs bad: {chat, chien, livre, garçon}

C Training Environment

Training ran both languages on cloud GPU infrastructure. Checkpoints saved every 1,000 steps.

D Training Logs

Real-time training logs available at: <https://github.com/adamzwasserman/fractal-language>

Component	Specification
Provider	Vast.ai
GPU	NVIDIA RTX 4090 (48 GB VRAM)
CUDA	13.0
CPU	32 cores / 137 GB RAM
Storage	Samsung NVMe (5.5 GB/s)
Framework	PyTorch
Cost	\$0.96/hr

Table 5: Training environment specifications.

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